

ILLESHEIM AHP, Germany

WMO: 107520

Lat: 49.477N

Lon: 10.382E

Elev: 329

StdP: 97.44

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-8.9	-15.2	1.0	-11.9	-10.9	1.5	-6.5	13.0	6.3	11.7	5.9	1.0	50	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	31.6	20.1	29.3	19.3	27.2	18.7	21.2	28.1	20.3	26.8	19.5	25.6	2.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.6	22.3	18.2	13.7	21.8	17.4	13.0	21.2	62.6	27.7	59.6	26.9	56.9	25.7	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.5	-14.2	34.4	4.5	1.9	-17.5	35.8	-20.1	36.9	-22.6	37.9	-25.8	39.3		
(5)			-15.5	23.1	4.0	1.4	-18.4	24.1	-20.7	24.9	-22.9	25.7	-25.8	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	0.4	1.0	5.0	9.0	13.1	17.0	19.0	18.6	14.3	9.6	5.1	2.0		
	DBStd	7.61	4.82	4.73	3.77	3.69	3.67	3.36	3.25	3.38	3.47	3.93	3.88	4.37		
	HDD10.0	1243	297	252	158	61	13	0	0	0	5	57	153	247		
	HDD18.3	3338	556	485	413	279	166	65	33	38	129	271	397	506		
	CDD10.0	1082	0	0	4	32	109	209	278	266	133	44	6	1		
	CDD18.3	134	0	0	0	0	4	24	52	46	7	1	0	0		
	CDH23.3	1729	0	0	0	11	72	345	644	549	102	5	0	0		
	CDH26.7	576	0	0	0	3	10	111	233	198	22	0	0	0		
(14) Wind	WSAvg	2.7	3.3	3.0	3.3	2.5	2.5	2.4	2.6	2.2	2.2	2.3	2.8	3.3		
(15) Precipitation	PrecAvg	716	52	50	51	42	66	73	79	62	57	61	61	61		
	PrecMax	955	113	122	141	117	157	155	149	138	125	201	129	170		
	PrecMin	481	10	11	8	4	19	27	26	19	17	18	3	11		
	PrecStd	108	28	29	32	25	35	33	35	30	27	34	27	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.8	19.8	24.8	28.5	32.7	34.9	34.2	29.4	24.0	16.8	12.2		
		MCWB	9.7	8.4	11.0	14.2	18.9	21.4	20.6	20.5	18.1	16.2	14.8	10.1		
	2%	DB	10.0	11.0	16.9	22.1	25.4	29.9	31.7	31.3	26.3	20.8	14.3	10.8		
		MCWB	8.7	8.0	10.1	13.0	17.4	19.9	20.2	19.9	17.8	15.1	12.2	9.4		
	5%	DB	8.1	9.2	14.2	20.0	23.2	27.2	29.2	28.8	23.8	18.1	12.8	9.3		
		MCWB	7.0	7.0	8.8	12.6	15.5	18.9	19.0	19.6	16.7	14.4	11.2	8.0		
	10%	DB	6.7	7.8	11.8	17.2	21.1	24.8	26.9	26.2	21.2	15.9	10.9	7.9		
		MCWB	5.8	5.8	8.3	11.5	14.9	17.9	18.4	18.4	15.8	13.7	9.6	7.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	9.9	12.3	15.9	20.2	23.0	22.1	22.1	19.9	17.4	15.2	11.1		
		MCDB	11.2	11.8	17.0	21.9	26.3	29.6	31.0	28.7	26.1	20.9	15.8	11.5		
	2%	WB	8.9	8.5	10.9	14.5	18.5	20.8	20.8	21.1	18.4	16.1	12.9	9.8		
		MCDB	9.7	10.5	15.2	20.2	23.6	27.3	27.8	28.0	24.0	19.2	14.2	10.6		
	5%	WB	7.2	7.4	9.6	13.2	17.0	19.6	20.0	20.2	17.5	15.0	11.4	8.2		
		MCDB	8.0	9.4	13.0	18.5	21.1	25.5	26.7	26.8	22.3	17.7	12.5	9.2		
	10%	WB	5.7	5.9	8.5	12.0	15.7	18.5	19.3	19.3	16.5	13.8	9.9	7.0		
		MCDB	6.6	7.4	11.2	17.0	20.0	23.9	25.3	24.9	20.4	15.7	10.8	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	7.3	10.1	12.2	11.7	11.9	12.1	12.2	11.2	9.2	6.1	5.4		
		MCDBR	6.4	10.4	16.2	17.7	15.6	16.4	17.3	16.2	15.5	13.1	9.2	7.0		
	5% WB	MCWBR	5.4	7.5	9.9	9.7	8.0	7.5	6.6	6.5	7.7	8.1	7.0	5.9		
		MCWBR	6.3	9.6	13.4	15.9	13.4	15.0	14.1	14.1	13.0	11.3	8.1	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.328	0.363	0.395	0.403	0.406	0.413	0.406	0.377	0.354	0.328	0.302		
		taud	2.429	2.375	2.316	2.254	2.260	2.281	2.257	2.298	2.360	2.419	2.445	2.427		
	Ebn at Noon		750	814	841	849	856	855	841	829	817	772	711	698		
		Edh at Noon	63	84	106	125	129	128	129	118	100	80	63	56		
(44) All-Sky Solar Radiation	RadAvg		0.85	1.67	2.78	4.26	5.03	5.58	5.28	4.58	3.34	1.98	0.96	0.67		
	RadStd		0.14	0.28	0.36	0.56	0.57	0.45	0.51	0.43	0.44	0.23	0.12	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page